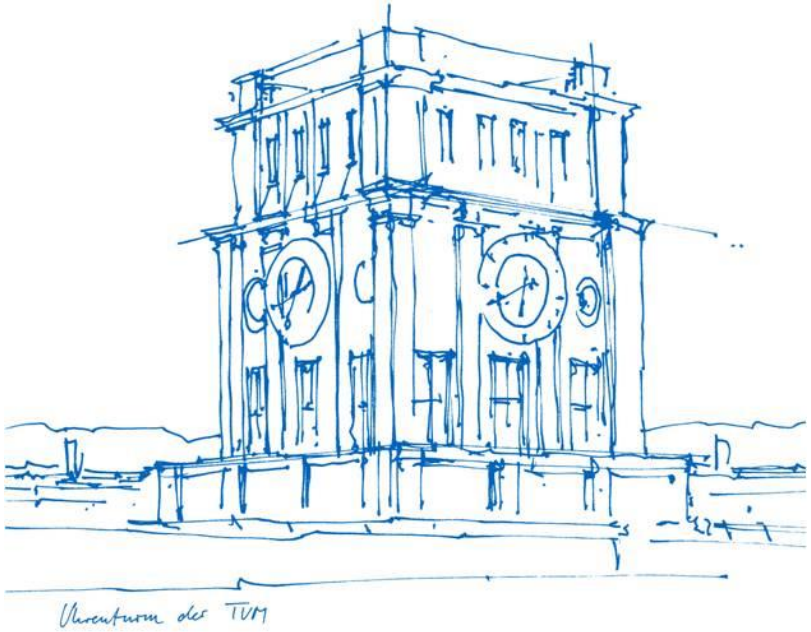
Grundlagenpraktikum: Rechnerarchitektur (GRA)

Maximilian Eichenseher, Timon Daniel, Hendrik Kahlsdorf

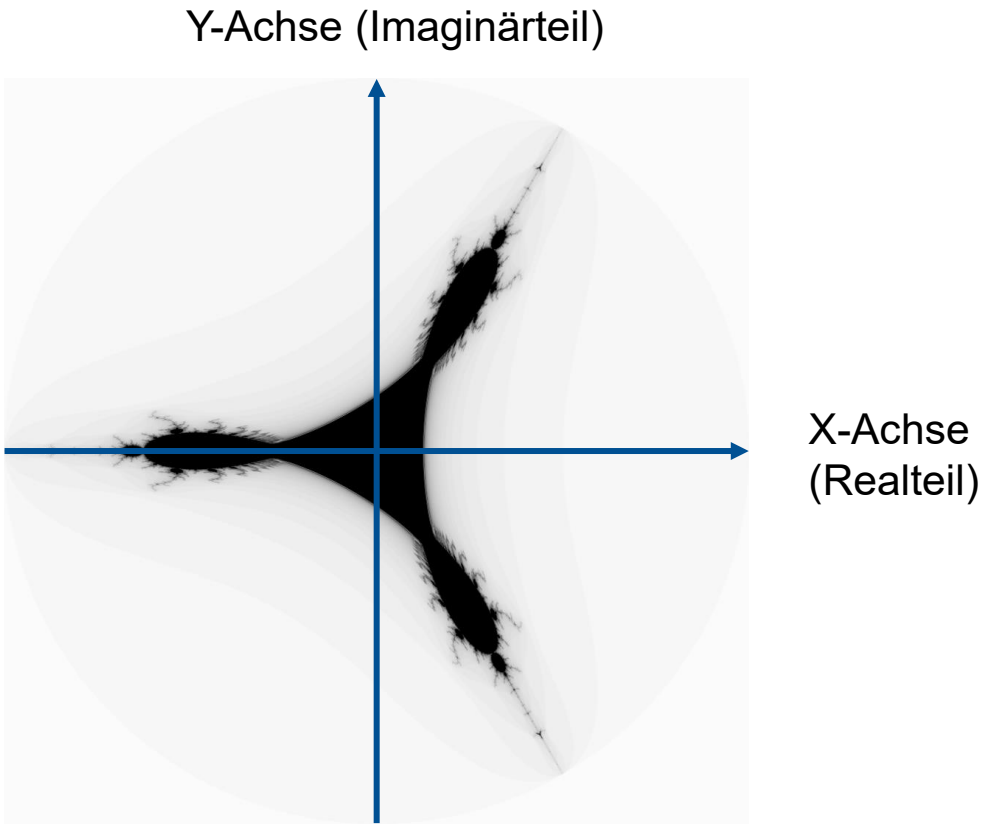
25. August 2026



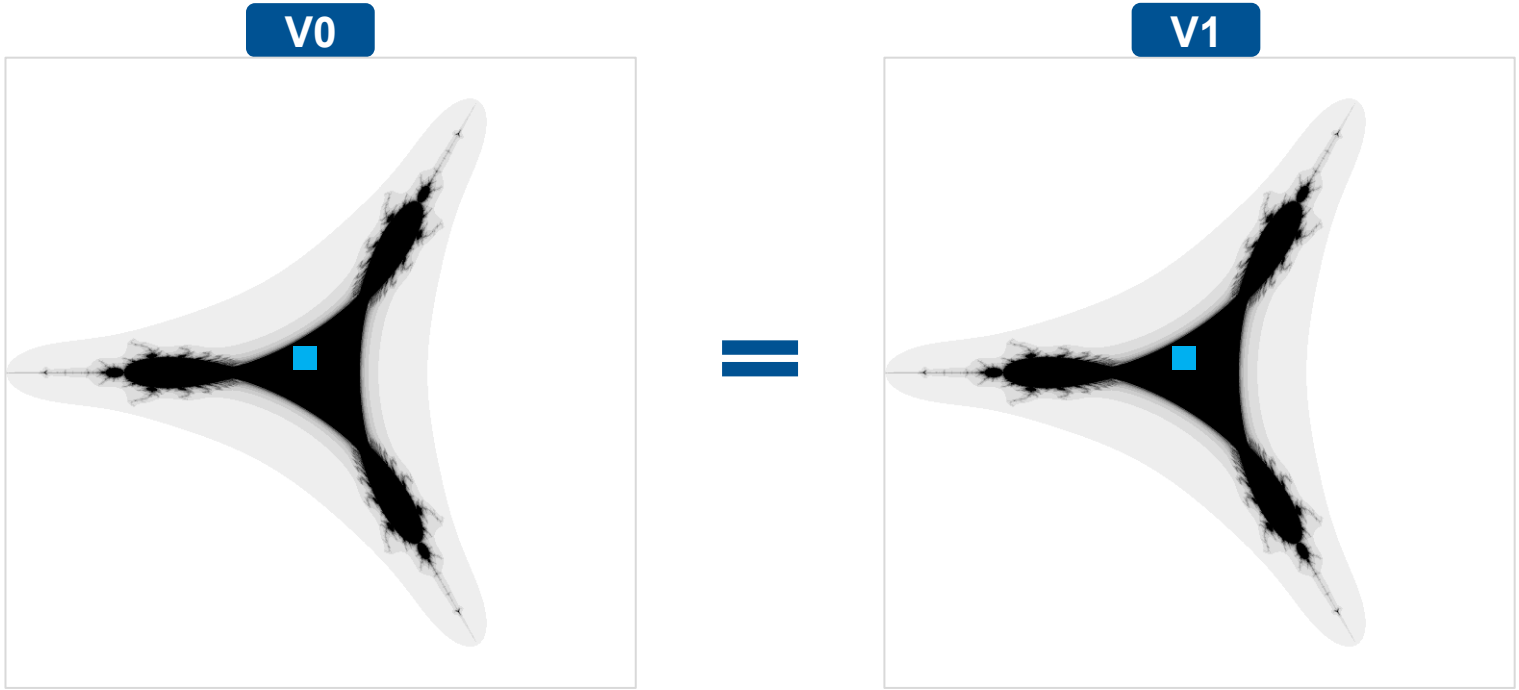
Erklärung Trikorn-Fraktal

Formel:

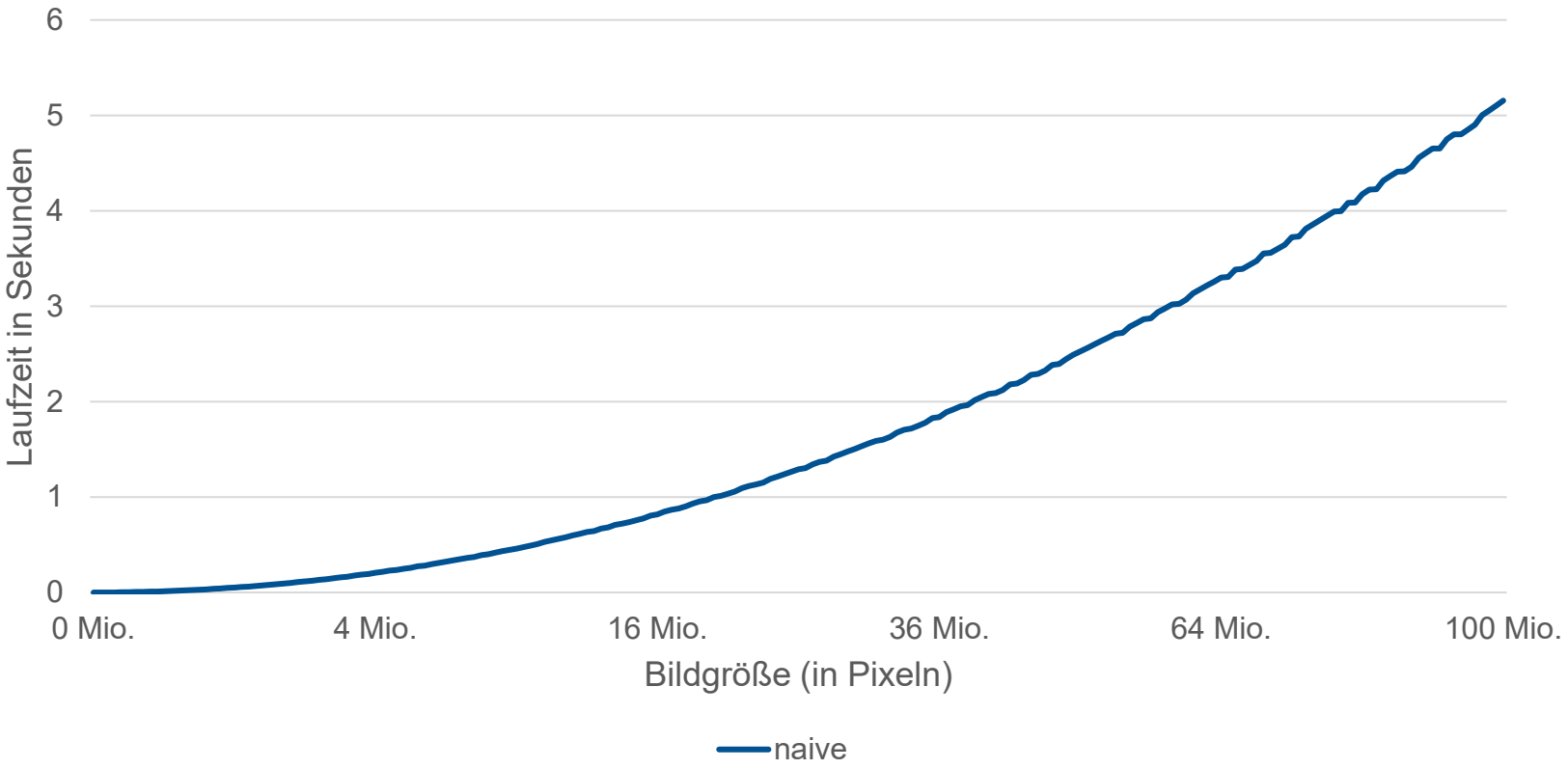
$$z_{i+1} = \overline{z_i}^m + c \; (i \geq 0)$$



Korrektheitsanalyse

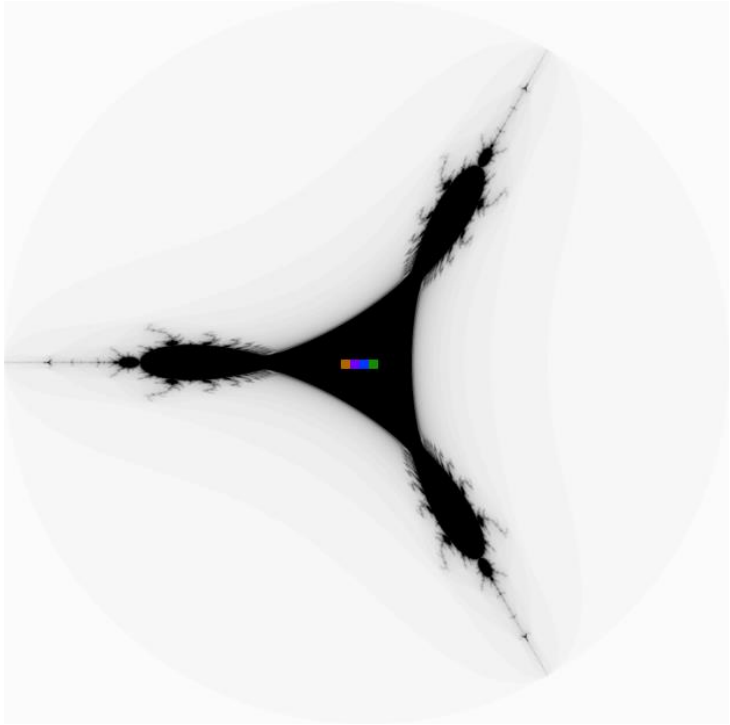


Laufzeit Naiv (Bildgröße)

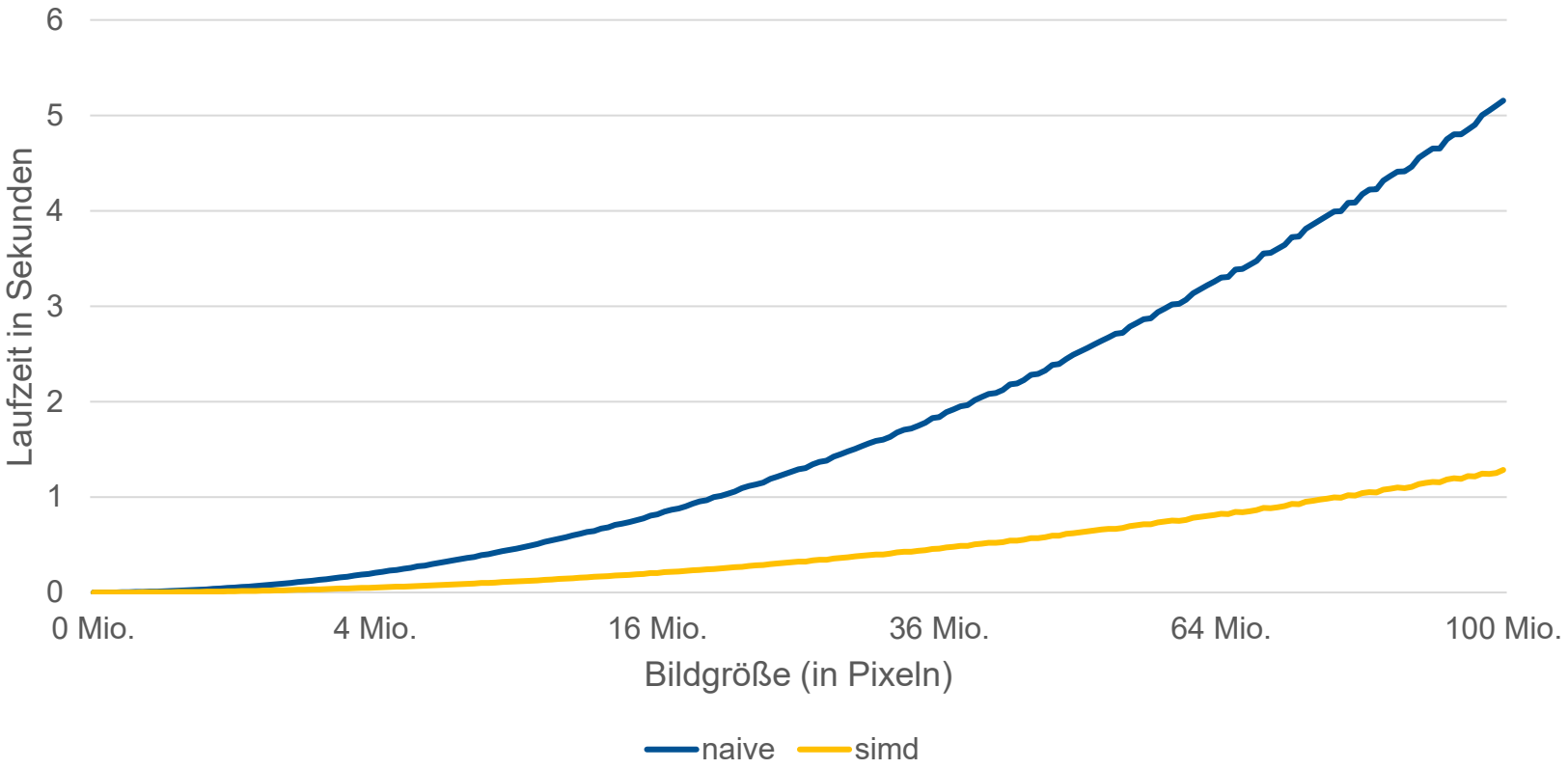


SIMD Optimierung

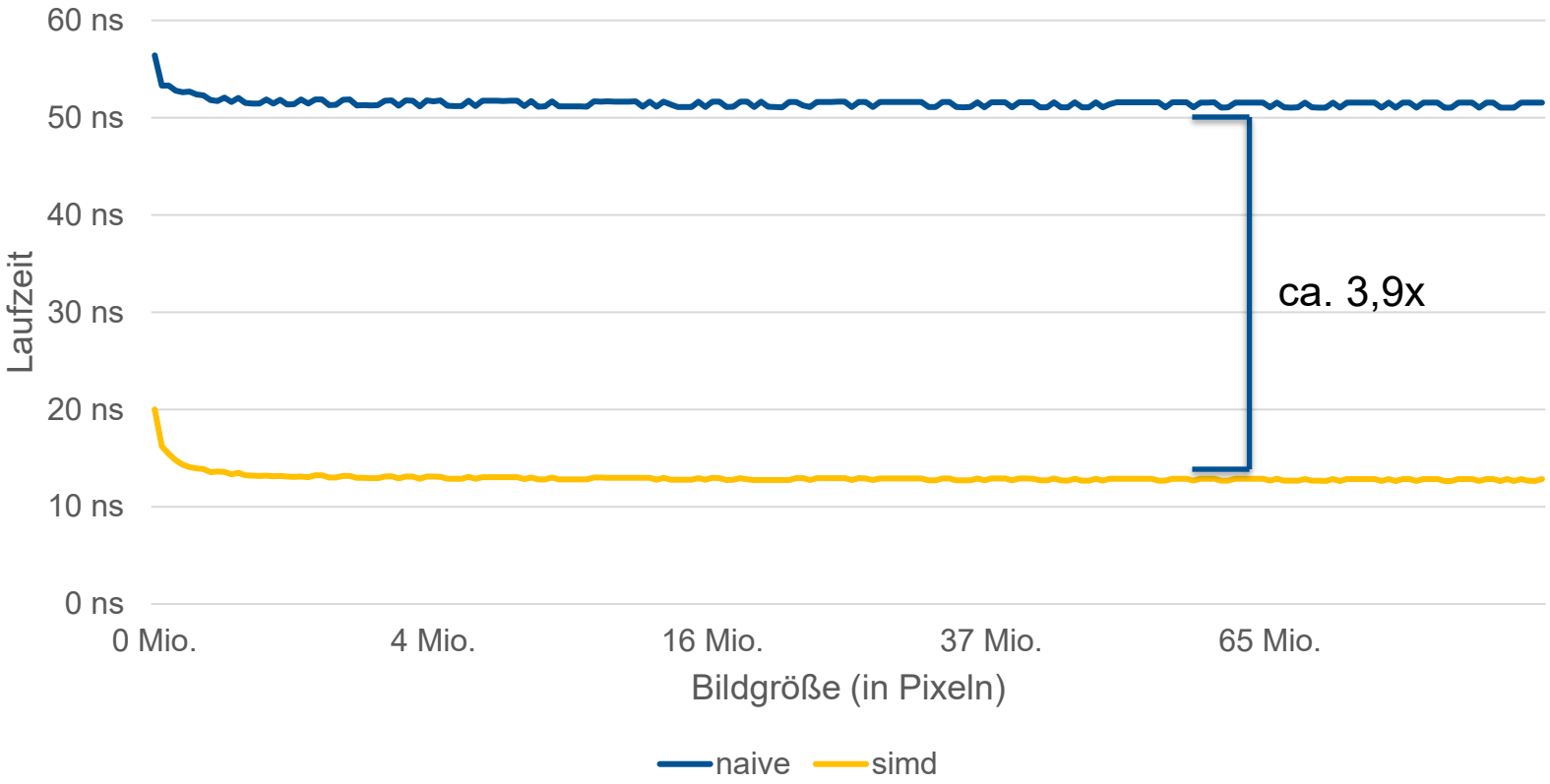
__m128 {x1, x2, x3, x4}



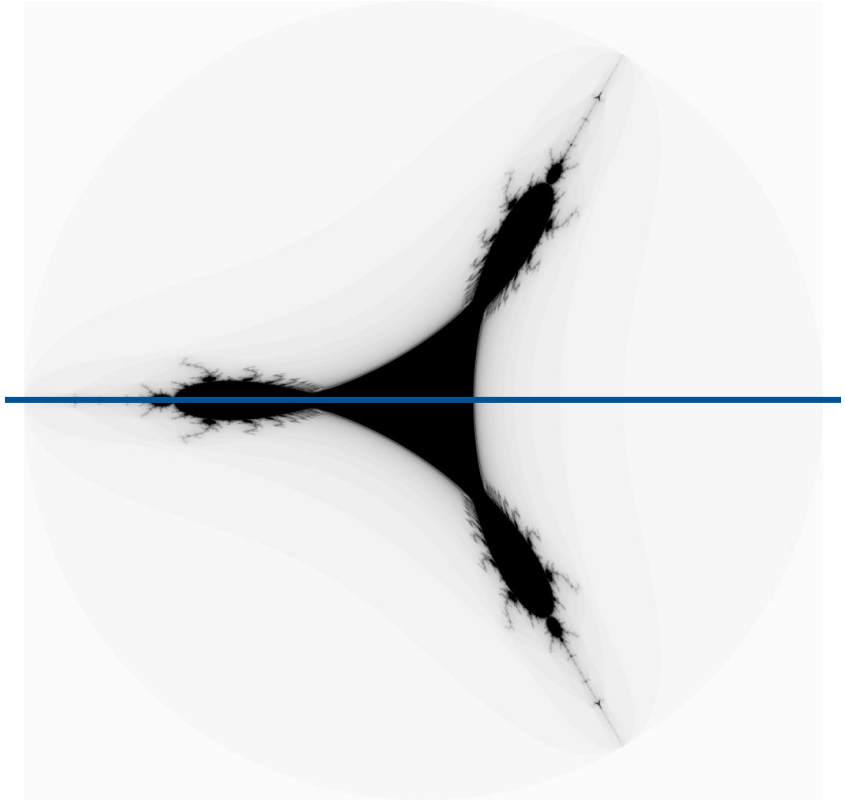
Laufzeit (Bildgröße): Naiv vs. Simd



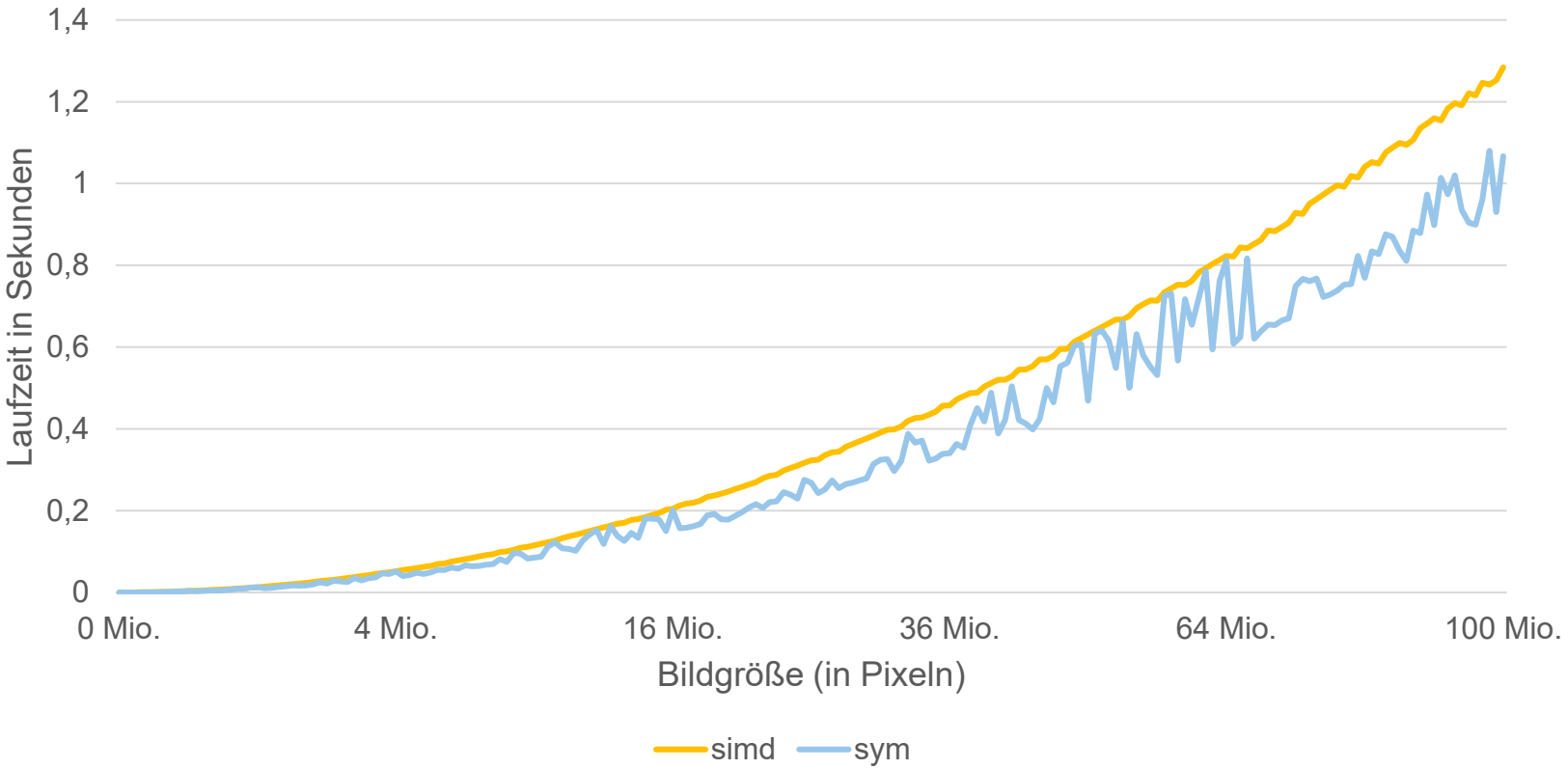
Laufzeit (pro Pixel): Naiv vs. SIMD



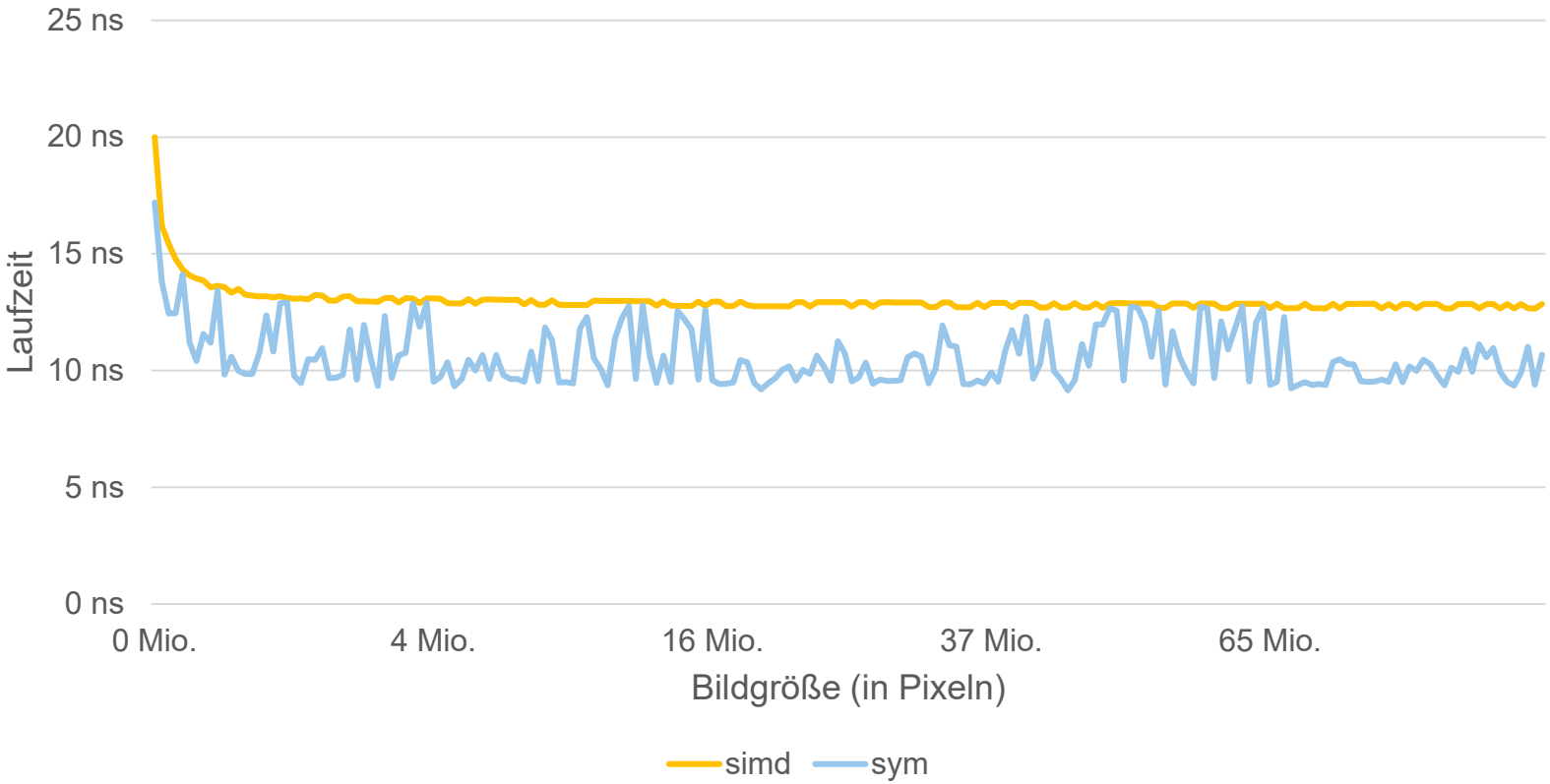
Symmetrie Optimierung



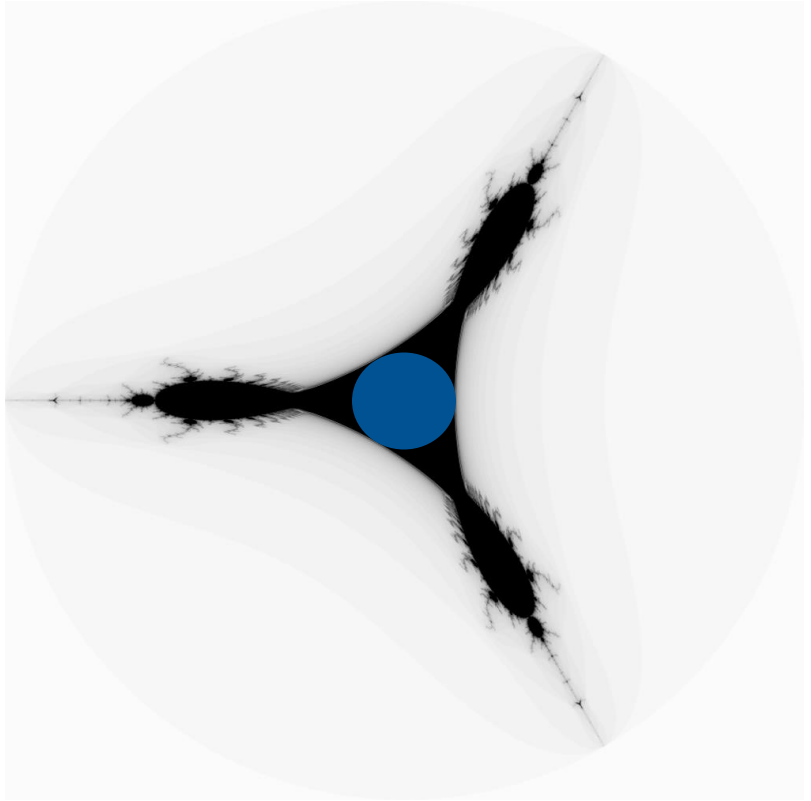
Laufzeit (Bildgröße): SIMD vs. Sym



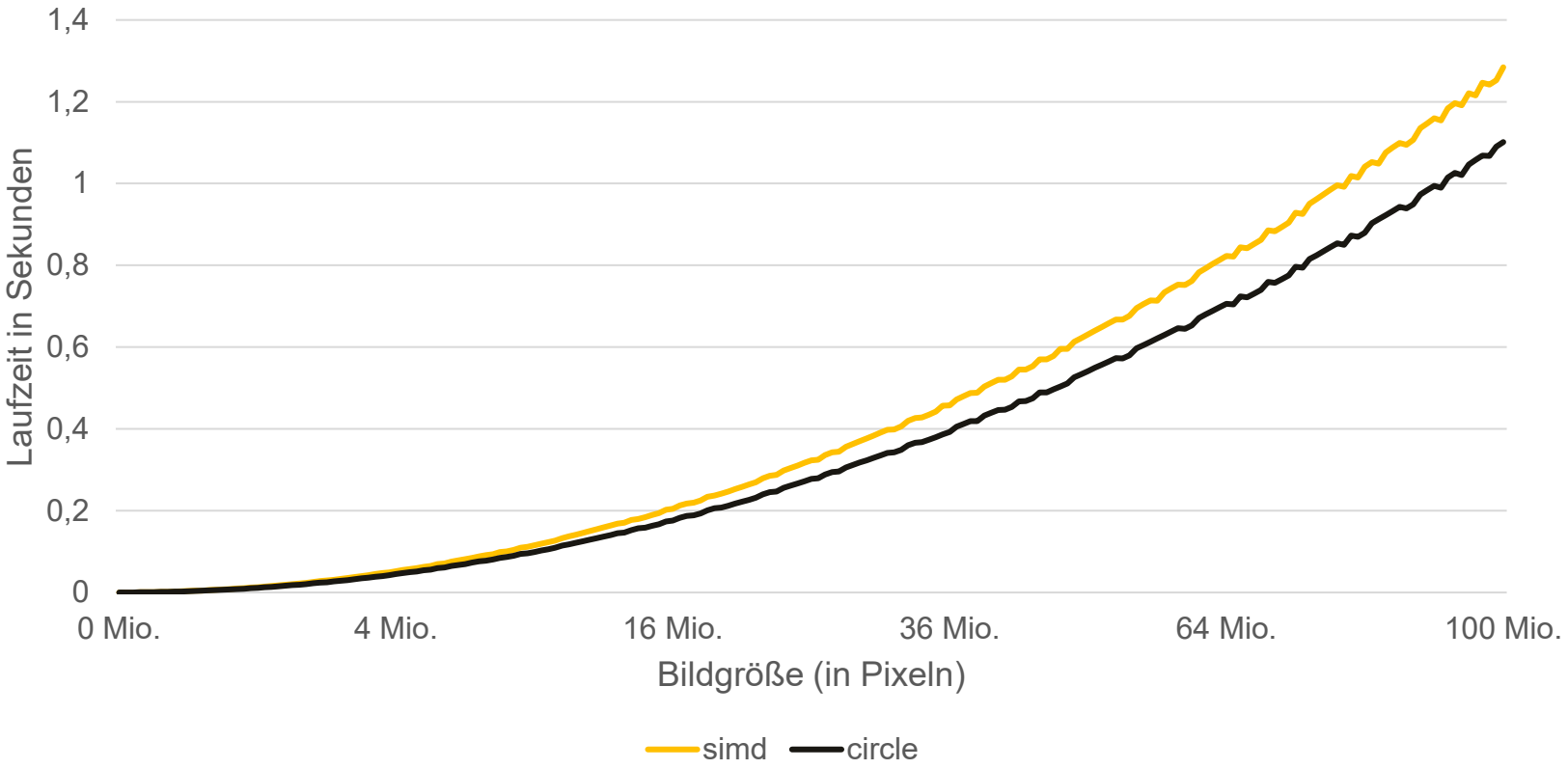
Laufzeit (pro Pixel): SIMD vs. Sym



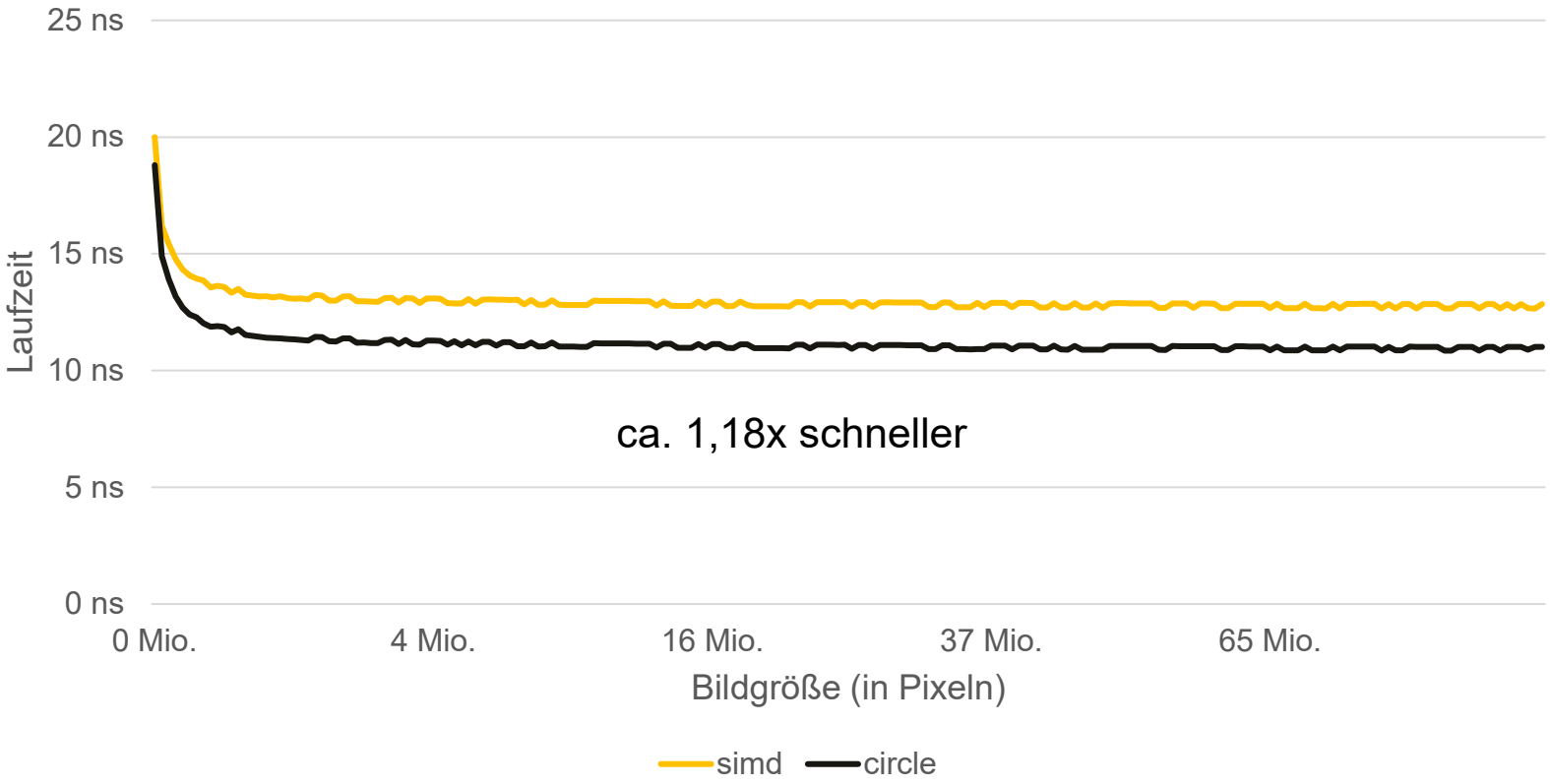
Circle Optimierung



Laufzeit (Bildgröße): SIMD vs. Circle



Laufzeit (pro Pixel): SIMD vs. Circle



Latency Throughput

_mm_add_ps (Intel Intrinsics)

Latency and Throughput

Architecture	Latency	Throughput (CPI)
Alderlake	-	0.5
Icelake Xeon	4	0.5
Sapphire Rapids	-	0.5
Skylake	4	0.5



8x Unterschied

Latency Throughput

```
x = _mm_add_ps (a, b)

_ = _mm_add_ps (_, x)
```

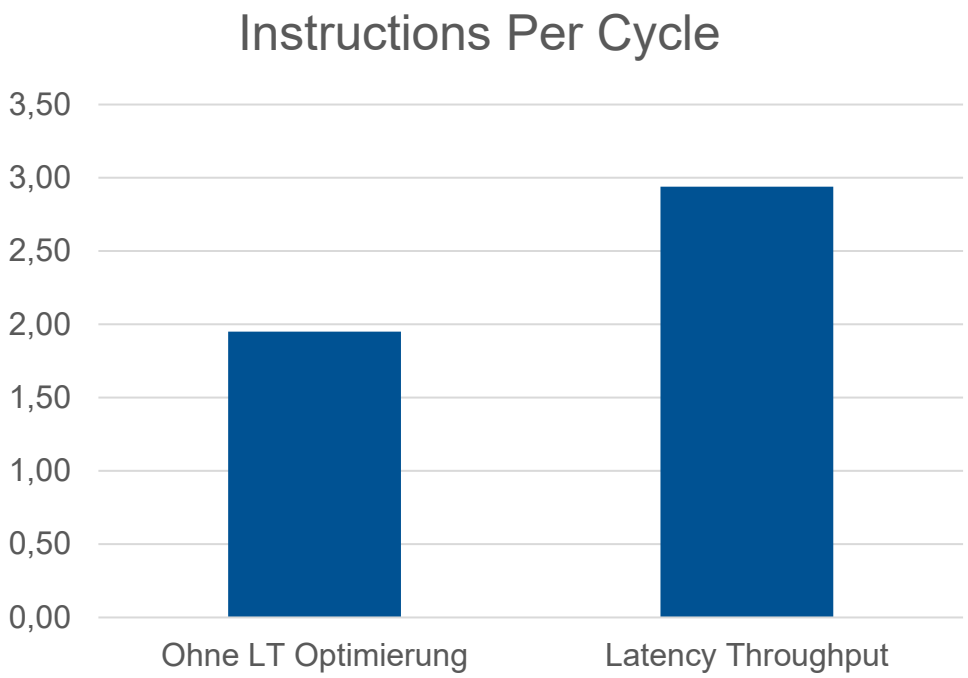


```
x = _mm_add_ps (a, b)
y = _mm_add_ps (c, d)

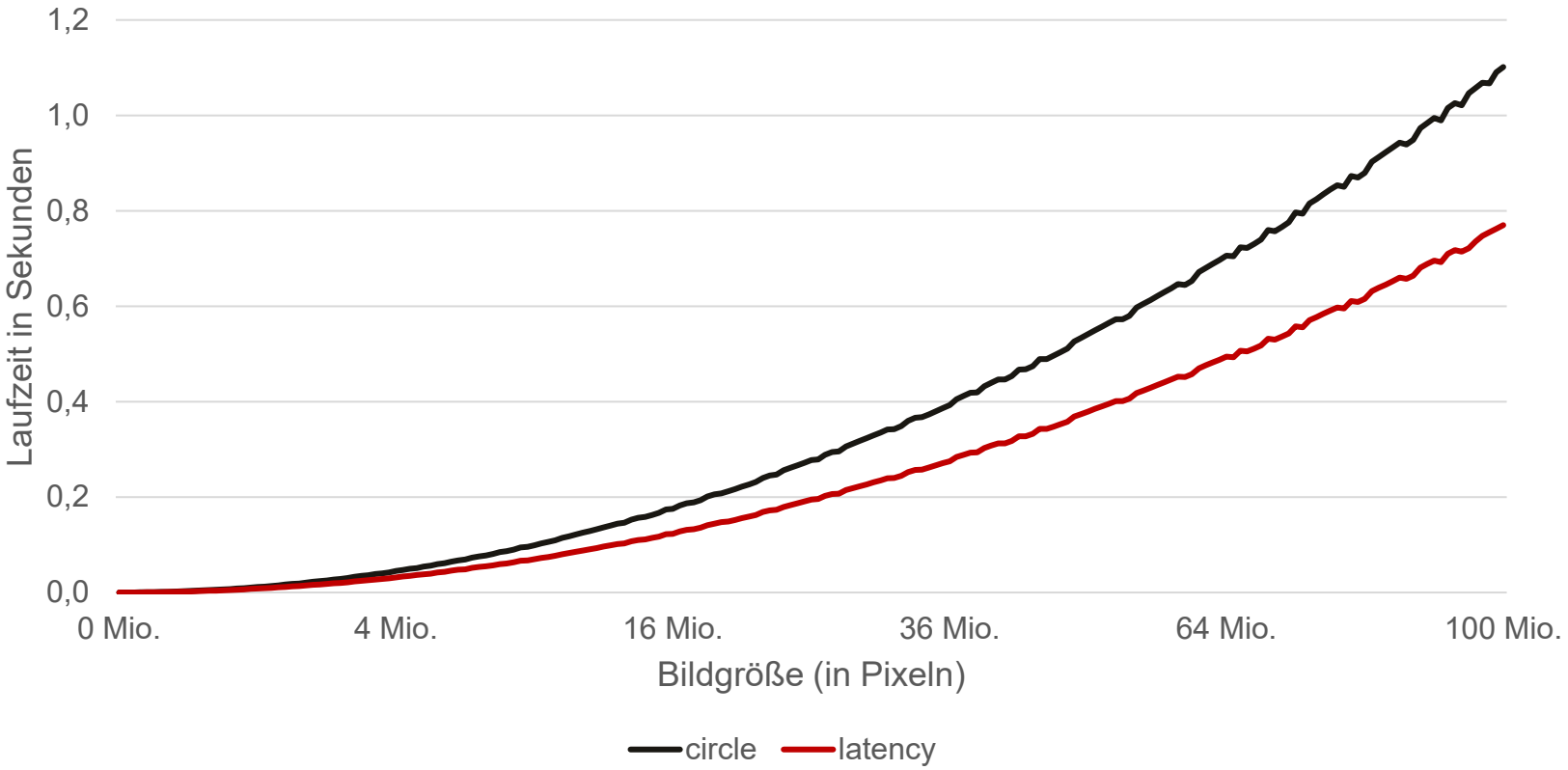
_ = _mm_add_ps (_, x)
_ = _mm_add_ps (_, y)
```

Latency Throughput

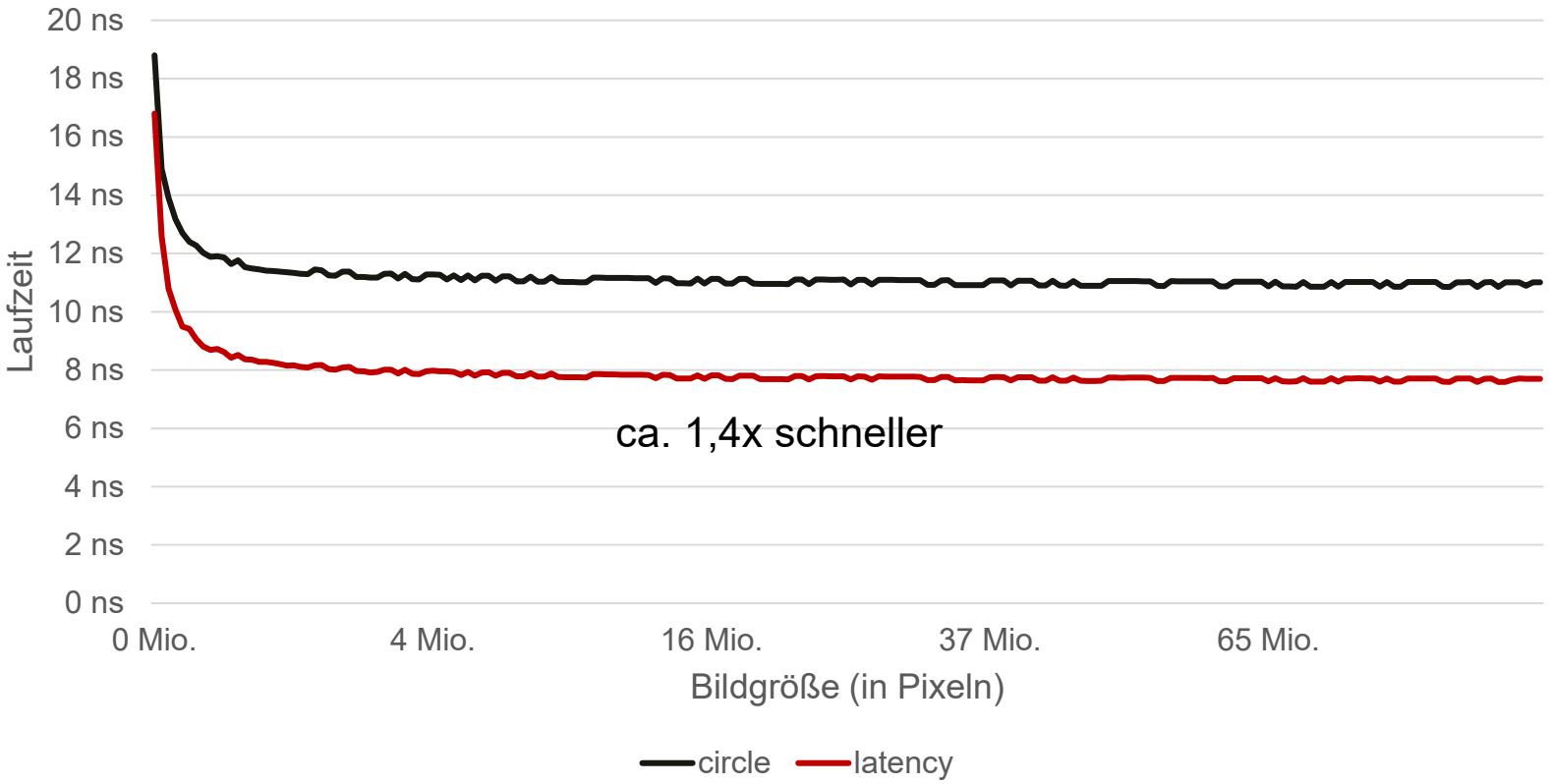
Perf Analyse



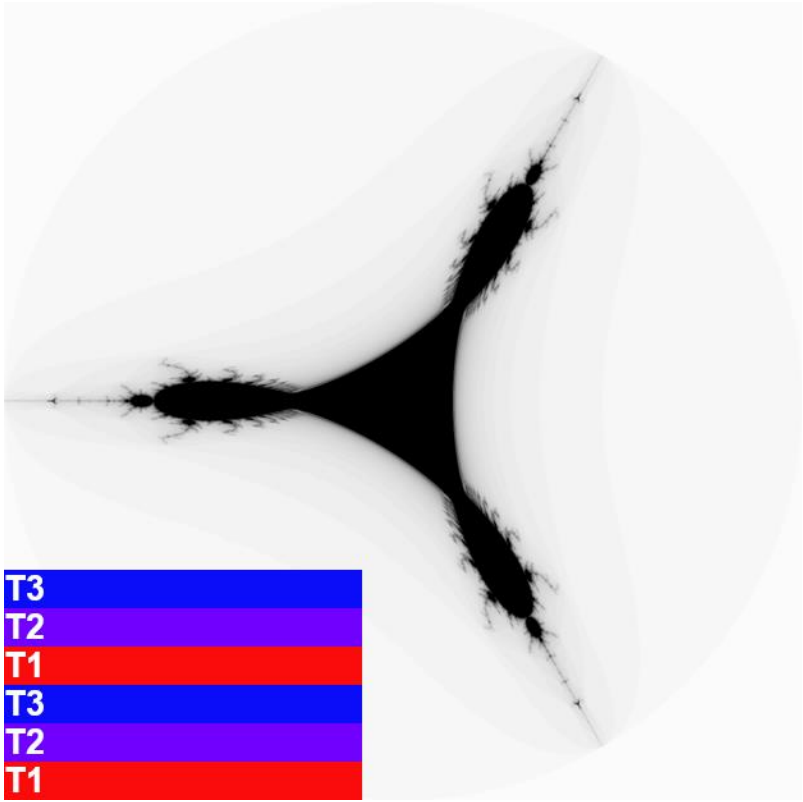
Laufzeit (Bildgröße): Circle vs. Latency



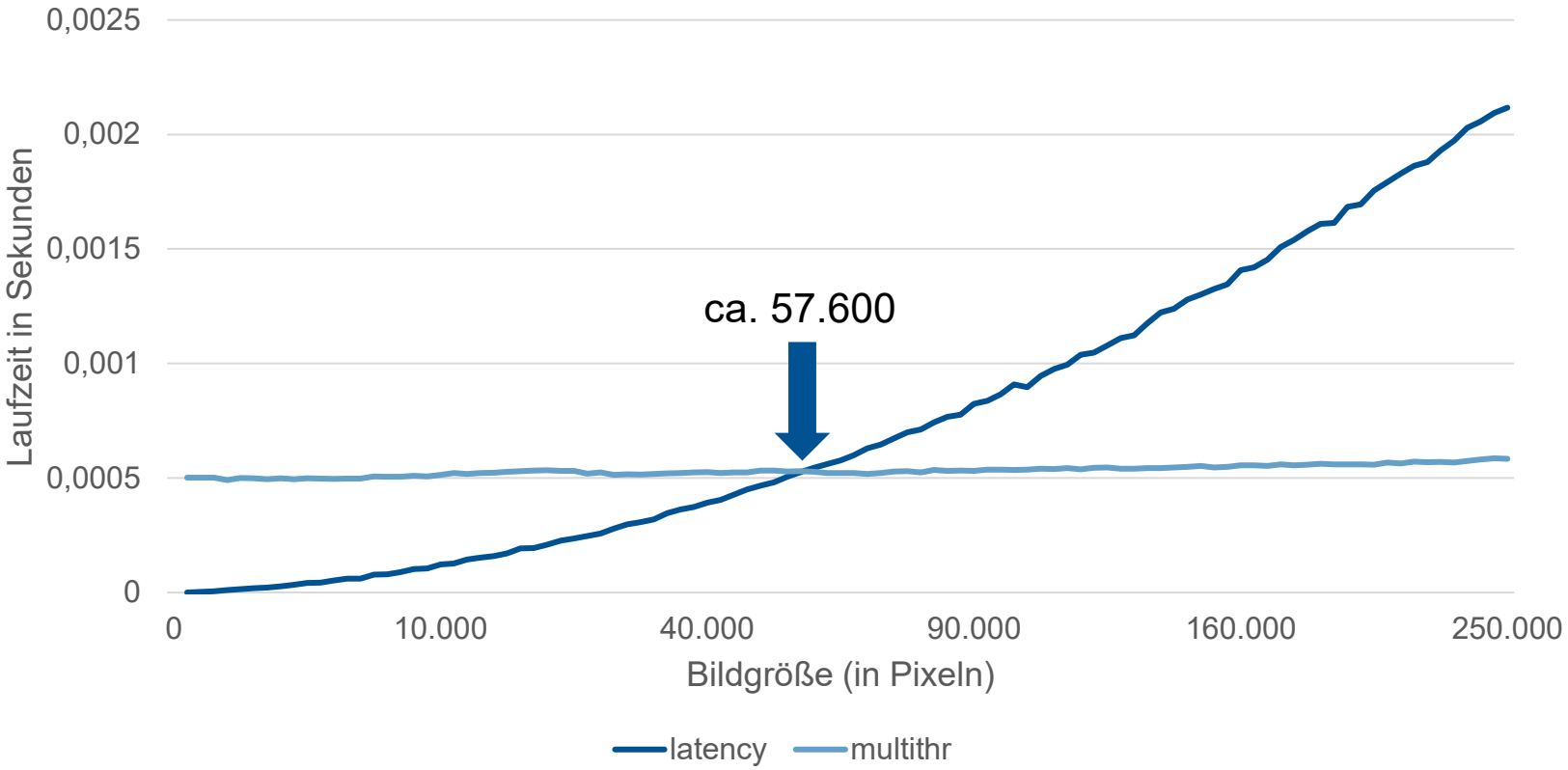
Laufzeit (pro Pixel): Circle vs. Latency



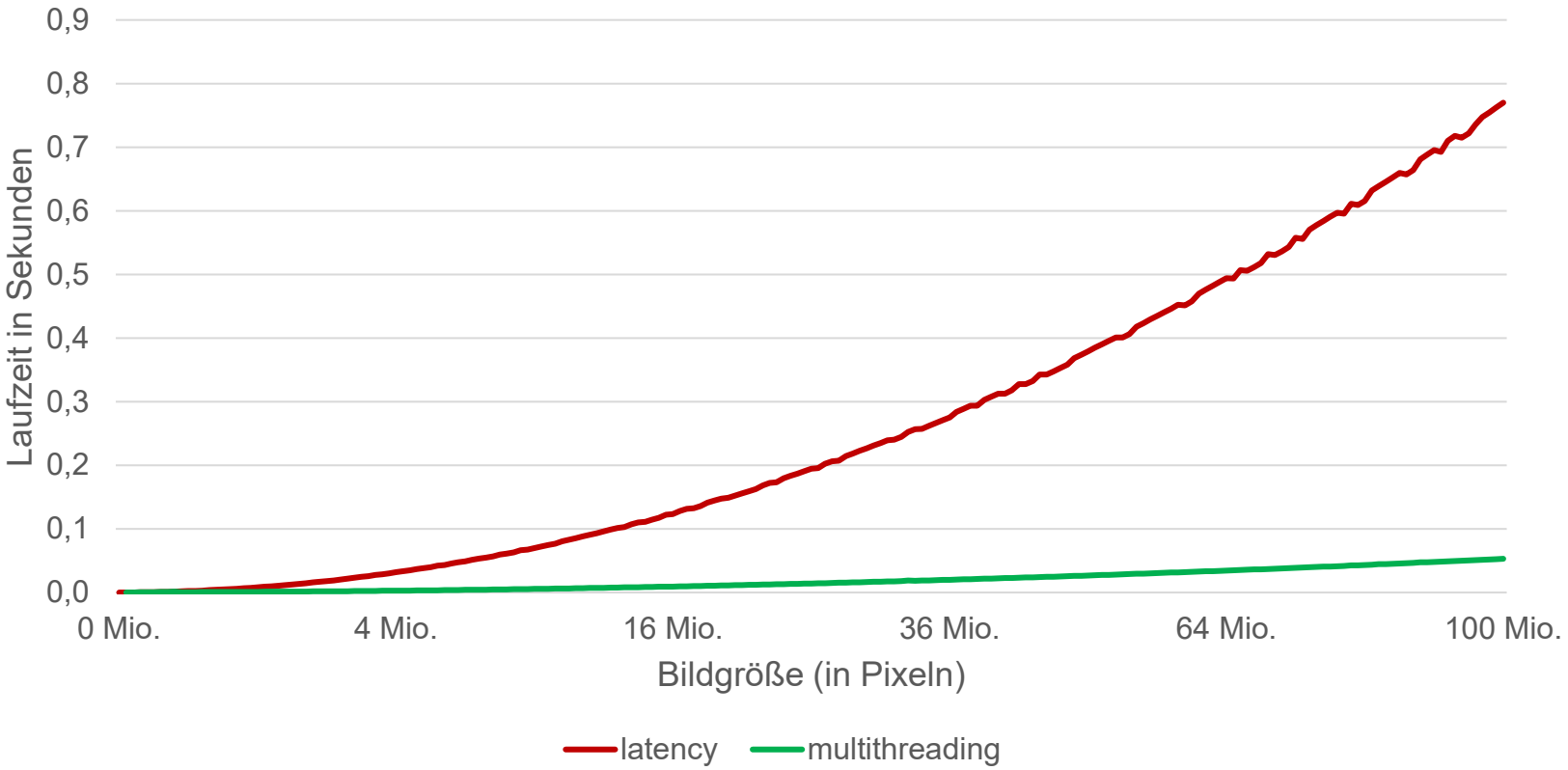
Multithreading



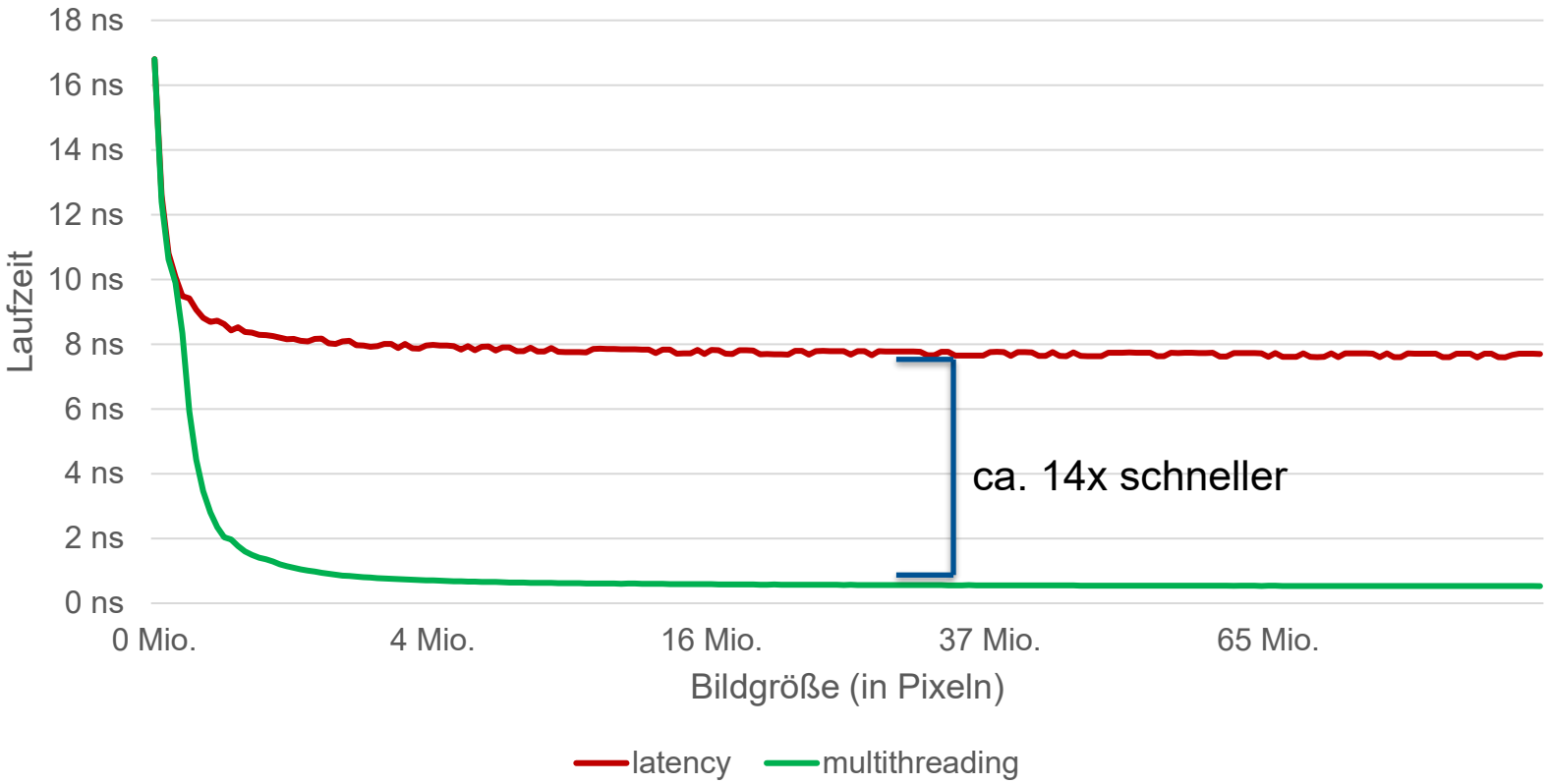
Thread-Overhead



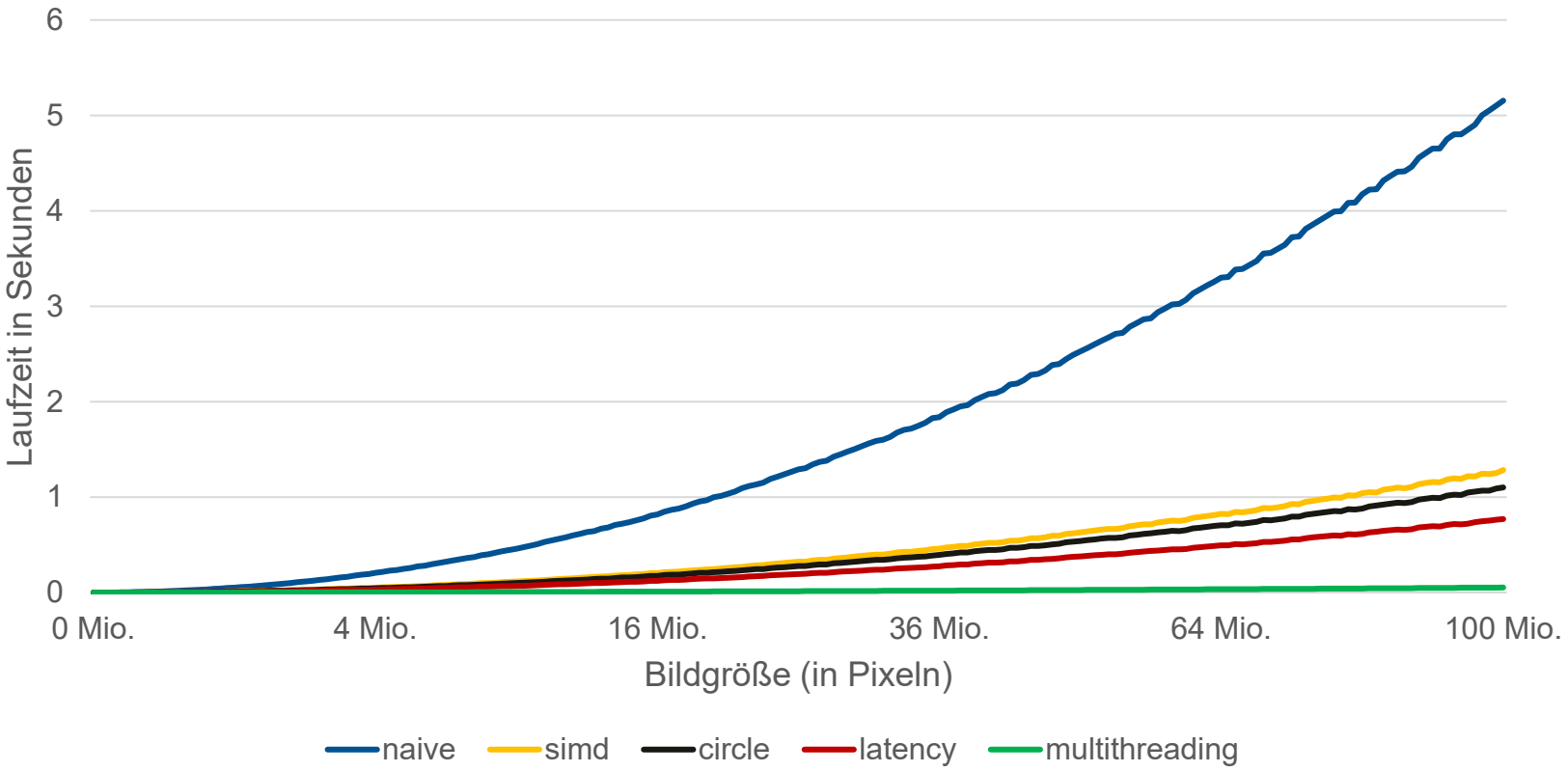
Laufzeit (Bildgröße): Latency vs. Threading



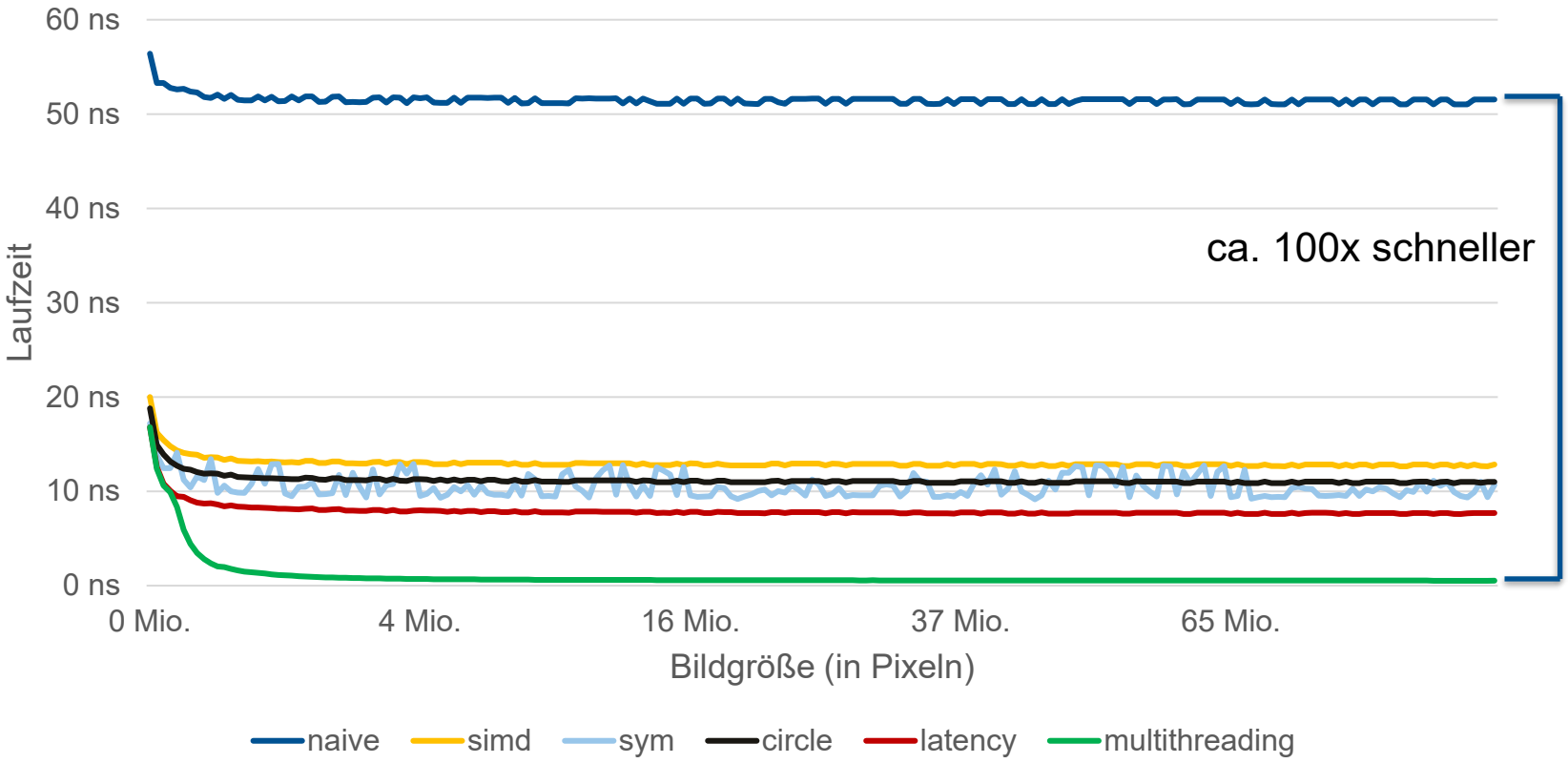
Laufzeit (pro Pixel): Latency vs. Threading

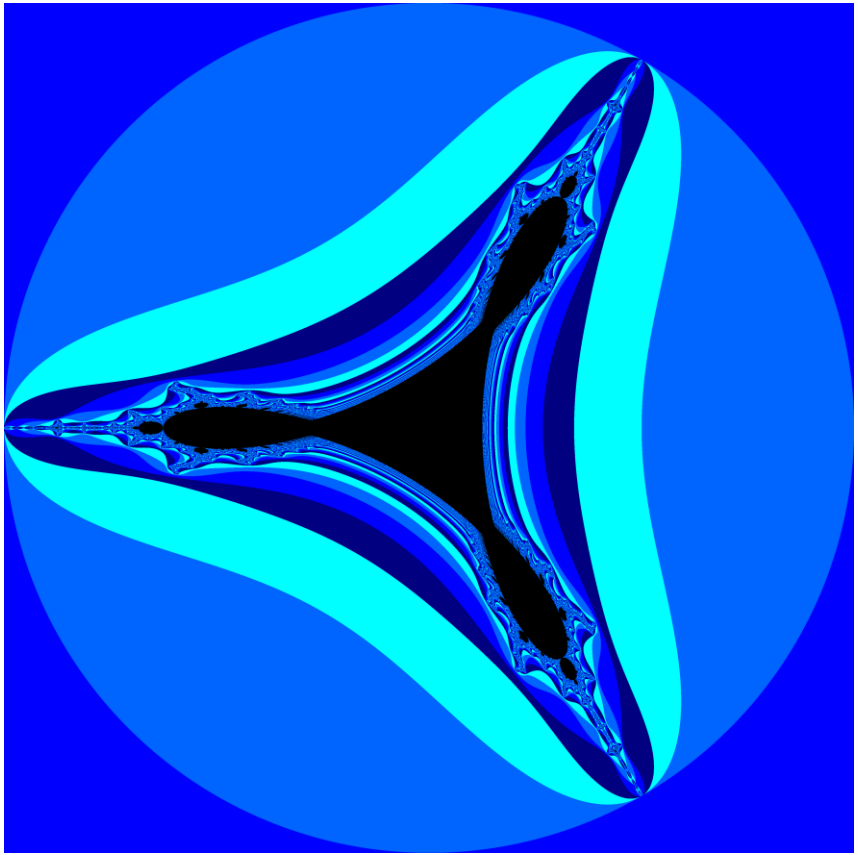


Laufzeit (Bildgröße)



Laufzeit (pro Pixel)





Testumgebung

- AMD Ryzen 9 5900X 12-Core Prozessor, 3.7GHz / 32 GB Arbeitsspeicher
- Ubuntu 26.04, 64 Bit, Linux-Kernel 7.0.0-14-generic
- Kompiliert mit GCC 15.2 mit der Option -O2
- quadratisches Bild mit Seitenlänge 50 bis 10.000 Pixeln (+50 je Durchgang)
- Resolution = $\frac{4}{Bildbreite}$
- Iterationen = 256
- Start = -2,-2
- beste Laufzeit aus drei Testdurchgängen